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COMMON QUARTERLY EXAMINATION - 2026

A

Standard XII

Reg.No. 1 2 B 0 0 9

MATHEMATICS

Time : 3.00 hrs **Part - A** *Marks : 90*

I. Choose the correct answer. **20 x 1 = 20**

1. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $\lambda A^{-1} = A$, then λ is

   a) 19 b) 14 c) 21 d) 17

2. If $\rho(A) = \rho(A\backslash B)$ then the system $AX = B$ of linear equation is

   a) consistent and has a unique solution b) consistent

   c) consistent and has infinitely many solutions d) inconsistent

3. The value of $\left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i}\right)^{10}$

   a) $\text{cis}\frac{2\pi}{3}$ b) $\text{cis}\frac{4\pi}{3}$ c) $-\text{cis}\frac{2\pi}{3}$ d) $-\text{cis}\frac{4\pi}{3}$

4. If z is a non zero complex number, such that $2iz^2 = \bar{z}$, then $|z|$ is

   a) $\frac{1}{2}$ b) 1 c) 2 d) 3

5. If $\left|z - \frac{3}{z}\right| = 2$, then the value of $|z|$ is

   a) 1 b) 2 c) 3 d) 5

6. The number of real numbers in $[0,2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1$ is

   a) 2 b) 4 c) 1 d) ∞

7. The polynomial $x^3 + 2x + 3$ has

   a) one negative and two imaginary zeros b) one positive and two imaginary zeros

   c) three real zeros d) no zeros

8. The domain of the function defined by $f(x) = \sin^{-1}\sqrt{x-1}$ is

   a) [1,2] b) [−1,1] c) [0,1] d) [−1,0]

9. If $\sin^{-1} x + \cot^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$ then x is equal to

   a) $\frac{1}{2}$ b) $\frac{1}{\sqrt{5}}$ c) $\frac{2}{\sqrt{5}}$ d) $\frac{\sqrt{3}}{2}$

10. If $f(x) = 0$ has n roots, then $f'(x) = 0$ has _____ roots.

    a) n b) n + 1 c) n − 1 d) n − 2

11. If P(x, y) be the point on $16x^2 + 25y^2 = 400$ with $F_1(3,0)$ and $F_2(-3,0)$, then $PF_1 + PF_2$ is

    a) 8 b) 6 c) 10 d) 12

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12 The eccentricity of the ellipse $(x-3)^2+(y-4)^2=\frac{y^2}{9}$ is

a) $\frac{1}{\sqrt{3}}$ b) $\frac{\sqrt{3}}{2}$ c) $\frac{1}{3\sqrt{2}}$ d) $\frac{1}{3}$

13 The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis is equal to half the distance between the foci is

a) $\frac{4}{3}$ b) $\frac{4}{\sqrt{3}}$ c) $\frac{2}{\sqrt{3}}$ d) $\frac{3}{2}$

14 If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}+\hat{j}$, $\vec{c}=\hat{i}$ and $(\vec{a}\times\vec{b})\times(\vec{c})=\lambda\vec{a}+\mu\vec{b}$ then the value of $\lambda+\mu$

a) 0 b) 1 c) 6 d) 3

15 $\vec{r}=s\hat{i}+t\hat{j}$ is the equation of (s, t are parameters)

a) ZOX plane b) a straight line joining points $\hat{i}$ and $\hat{j}$ c) XOY plane d) YOZ plane

16 If $\vec{a}$ and $\vec{b}$ are unit vector such that $[\vec{a},\vec{b},\vec{a}\times\vec{b}]=\frac{1}{4}$ then the angle between $\vec{a}$ and $\vec{b}$ is

a) $\pi/6$ b) $\pi/4$ c) $\pi/3$ d) $\pi/2$

17. If the distance of the point (1,1,1) from the origin is half of its distance from the plane $x+y+z+k=0$ then the value of K are

a) ±3 b) ±6 c) −3,9 d) 3,−9

18 If A is a square matrix of order m x n, then ρ(A) is

a) m b) n c) ≤ max(m,n) d) ≤ min(m,n)

19 If $4\sin^{-1}x+\cos^{-1}x=\pi$, then x is equal to

a) $3/2$ b) $1/\sqrt{2}$ c) $2/\sqrt{3}$ d) $\sqrt{3}/2$

20 When the eccentricity of the ellipse become zero, then it becomes a

a) straight line b) Parabola c) point d) circle

Part - B

II. Answer any Seven questions. (Q.No.30 is compulsory) 7 x 2 = 14

21. Prove that $\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$ is orthogonal

22. Find the inverse of the matrix $\begin{bmatrix}2 & -1\\ 5 & -2\end{bmatrix}$ by Gauss–Jordan method

23. If $|z|=2$, Prove that $3\le|Z+3+4i|\le 7$

24. If $(x_1+iy_1)(x_2+iy_2)(x_3+iy_3)\ldots\ldots(x_n+iy_n)=a+ib$, show that $(x_1^2+y_1^2)(x_2^2+y_2^2)(x_3^2+y_3^2)\ldots\ldots(x_n^2+y_n^2)=a^2+b^2$

25 Form a polynomial equation with integer coefficients with $\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ as a root.

26 Find the domain of $g(x)=2\sin^{-1}(2x-1)-\frac{\pi}{4}$

27 Find the equations of the tangent to the hyperbola at $12x^2-9y^2=108$ at $\theta=\pi/3$

28. A particle acted on by constant forces $8\hat{i}+2\hat{j}-6\hat{k}$ and $6\hat{i}-2\hat{j}-2\hat{k}$ is displaced from the point (1,2,3) to the point (5,4,1) Find the total work done by the forces

29. Find the intersects cut off the plane $\vec{r}\cdot(6\hat{i}+4\hat{j}-3\hat{k})=12$ on the co–ordinate axis.

30. Prove that $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{7}=\tan^{-1}\frac{9}{13}$

Part - C

III. Answer any Seven questions. (Q.No.40 is compulsory) 7 x 3 = 21

31. Find the rank of the matrix $\begin{bmatrix}1 & 2 & 3\\ 2 & 1 & 4\\ 3 & 0 & 5\end{bmatrix}$ by reducing if to a row–echelon form.

32. Solve the system of equation by inverse method $2x+5y+2=0$, $x+2y=-3$

33. Find the square root of 6–8i

34. Which one of the point 10– 8i, 11+6i is closest to 1+i

35. Find a polynomial equation of minimum degree with rational coefficients, having 2i+3 as a root.

36. Find the value of $\cos^{-1}\left(\frac{1}{2}\right)+\sin^{-1}(-1)$

37. The minimum and maximum distances of the Earth from the sun respectively are 152×10^6 km and 94.5×10^6 km. The sun is at one focus of the elliptical orbit. Find the distance from the sun to the other focus.

38. Prove that $[\vec{a}-\vec{b},\vec{b}-\vec{c},\vec{c}-\vec{a}]=0$

39. Prove by vector method that an angle in a semi–circle is a right angle

40. Find the equation of the circle through the points (3,0), (−3,0) (0,3) and (0,−3)

Part - D

IV. Answer all the questions. 7 x 5 = 35

41. a) Solve the following systems of linear equations by Cramer's rule
$3x+3y-z=11$, $2x-y+2z=9$, $4x+3y+2z=25$

(OR)

b) Find the parametric form of vector equation, and Cartesian equations of the plane containing the line $\vec{r}=(\hat{i}-\hat{j}+3\hat{k})+t(2\hat{i}-\hat{j}+4\hat{k})$ and perpendicular to plane $\vec{r}\cdot(\hat{i}+2\hat{j}+\hat{k})=8$

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42. a) Let Z_1, Z_2 and Z_3 are three complex numbers such that $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|z_1 + z_2 + z_3| = 1$, show that $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 6$

(OR)

b) Identify the type of conic and find centre, foci, vertices, and directrices of $9x^2 - y^2 - 36x - 6y + 18 = 0$

43. a) Solve the equation: $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution.

(OR)

b) Solve : $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$

44. a) If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ and $0 < x,y,z < 1$ show that $x^2 + y^2 + z^2 + 2xyz = 1$

(OR)

b) Find all zeros of the polynomial $x^6 - 3x^5 - 5x^4 + 22x^3 - 39x^2 - 39x + 135$, if it is known that $1 + 2i$ and $\sqrt{3}$ are two of its zeros.

45. a) On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of 4m when it is 6m away from the point of projection. Finally it reaches the ground 12m away from the starting point. Find the angle of projection.

(OR)

b) Solve the equation $z^3 + 27 = 0$

46. a) Prove by vector method that $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

(OR)

b) Investigate the values of λ and μ the system of linear equations

$2x + 3y + 5z = 9$, $7x + 3y - 5z = 8$, $2x + 3y + \lambda z = \mu$

have

(i) No solution

(ii) A unique solution

(iii) An infinite number of solutions.

47. a) If $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{c} = 3\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{d} = 2\hat{i} + 5\hat{j} + \hat{k}$

verify that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a}, \vec{b}, \vec{d}]\vec{c} - [\vec{a}, \vec{b}, \vec{c}]\vec{d}$

(OR)

b) Show that the line $x - 9y + 13 = 0$ is a tangent to the ellipse $2x^2 + 7y^2 = 14$. Also find the coordinates of the point of contact.

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